Paper G-01: Language Evolution: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Shared Part IV notation and evidence boundaries are defined in `PART_IV_METHODS_AND_CLAIM_BOUNDARY.md`*. This paper states only its local observables, comparison, and failure conditions.*

Abstract

This paper develops a falsifiable CDFD/AFL model of Language Evolution. It maps utterances, meanings, learners, media, and social interaction to driving flux Φ, cognition, channel capacity, social structure, ambiguity, and convention to active constraint C, innovation, repair, learning, borrowing, and accommodation to responsiveness S, and grammar, lexicon, corpora, identity, institutions, and historical contact to structural memory M_s. The target outcome is intelligibility, change, diffusion, and linguistic diversity, tested against corpora, experiments, social networks, historical records, and longitudinal language data. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

1 Introduction

Language is not a static repository of symbols; it is an active, self-regulating biological and social phenomenon. Words are invented, adopted, mutated, and discarded at rates that mirror genetic drift and viral epidemiology.

When a language spreads across a wide geographic and social space, changing contact networks can limit communication and support dialect formation. CDFD treats this as a testable interaction among information flow, network constraint, adaptive repair, and historical memory rather than an inevitable breakdown.

2 Linguistics as an Adaptive Surface

We apply the Adaptive Operating Ratio to the domain of human communication:

$$\Psi_s = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s \tag{1}$$

For a language group, the parameters map directly to informational transport:

- **Flux (Φ)**: The daily volume of communication, trade, and memetic exchange occurring across the population.
- **Constraint (C)**: Geographic boundaries (mountains, oceans), political borders, or simply the physical limits of ancient communication speeds.
- **Responsiveness (S)**: The cognitive plasticity of the population, particularly the youth, who act as the primary agents of linguistic innovation and adoption.
- **Memory (M_s)**: The formal grammar, written records, and academic institutions that standardize the language. High memory rigidly enforces "correct" usage, resisting phonetic drift.

2.1 Dialectic Bifurcation and Regime Shifts

Consider a unified language spoken across an expanding empire. Initially, trade and travel are high (Φ is large), maintaining a healthy communicative regime ($\Psi_s \approx 1$).

If the empire fractures, the constraint (C) between two provinces drastically increases. The cross-border flux Φ drops. Consequently, the adaptive ratio across that specific geographical border plummets to $\Psi_s \ll 1$. The system is in "communicative decay."

To restore local equilibrium, the isolated nodes (provinces) must adapt internally. Cut off from the central normalizing flux, their local youth populations (high S) begin optimizing the language for local efficiency, introducing slang and phonetic shifts. The central, formal grammar (high M_s) can no longer suppress these changes because the regulatory flux (Φ) from the capital is gone. The language bifurcates. What was once a unified active surface topologically splits into two distinct, localized surfaces, each achieving its own stable $\Psi_s \approx 1$ state as a new, distinct language (e.g., the fracturing of Latin into the Romance languages).

3 Measurement Layer

4 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

4.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Language Evolution, the proposed drive is utterances, meanings, learners, media, and social interaction. The active limitation is cognition, channel capacity, social structure, ambiguity, and convention. Responsiveness is carried by innovation, repair, learning, borrowing, and accommodation, while retained state is represented by grammar, lexicon, corpora, identity, institutions, and historical contact. The outcome to explain is intelligibility, change, diffusion, and linguistic diversity.

Table 1: Operational CDFD/AFL map for Paper G-01.

Term	**Paper-specific observable meaning**	**Measurement question**
Φ	utterances, meanings, learners, media, and social interaction	What rate, load, gradient, or demand enters the focal system?
C	cognition, channel capacity, social structure, ambiguity, and convention	Which measured bottleneck limits transfer, function, or recovery?
S	innovation, repair, learning, borrowing, and accommodation	Which process changes effective capacity or reroutes the load?
M_s	grammar, lexicon, corpora, identity, institutions, and historical contact	Which retained state makes equal present loads produce different futures?
Y	intelligibility, change, diffusion, and linguistic diversity	Which outcome is predicted out of sample, with uncertainty?

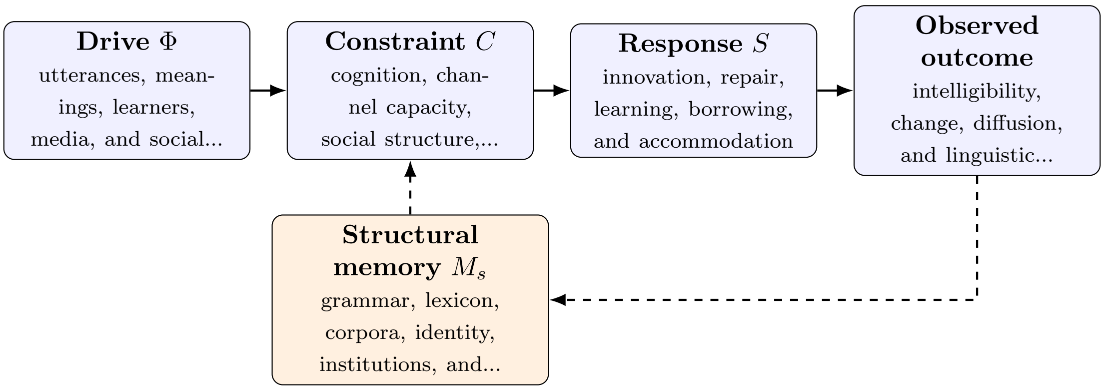

Figure 1: Paper G-01 observable CDFD/AFL architecture for Language Evolution. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

4.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in utterances, meanings, learners, media, and social interaction arrives faster than innovation, repair, learning, borrowing, and accommodation can compensate. This raises or redistributes cognition, channel capacity, social structure, ambiguity, and convention. If the load persists, the retained state encoded by grammar, lexicon, corpora, identity, institutions, and historical contact changes the next response, so a later event of equal magnitude can produce a different trajectory in intelligibility, change, diffusion, and linguistic diversity. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

4.3 Cross-Scale Translation Without Mechanistic Erasure

For the abstract and cognitive systems family, the cross-domain translation compares information-bearing activity, limited channels or institutions, adaptive reorganization, and retained semantic or technical structure. Because meanings and goals are not conserved physical quantities, every mapping must state its observational unit and avoid treating metaphor as measurement. [3, 2, 5, 4, 1] The required scale declaration for this paper is milliseconds to millennia; speakers to language families. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

4.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper G-01. Each rung can fail independently.

Test	**Design**	**Discriminating result**
Proxy audit	Measure corpora, experiments, social networks, historical records, and longitudinal language data and preregister how each record maps to Φ, C, S, M_s, and Y.	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a communication or contact shift across communities with different histories while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper's stated falsifier is: constraint-memory terms do not improve language-change or comprehension prediction.

4.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from corpora, experiments, social networks, historical records, and longitudinal language data and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of intelligibility, change, diffusion, and linguistic diversity. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

5 Conclusion

Language evolution is not exempt from the laws of universal thermodynamics. Grammatical rules are systemic memory; communication is energetic flux; and dialects are structural bifurcations born of chronic constraint. CDFD suggests that information networks, whether composed of neurons, electrical grids, or human speech, optimize their topology using the same candidate mathematical structure.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

6 Reference Layer

References

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